MIT END SEMESTER EXAMINATION

Course name: Analog Circuits (ECE 2126), Max Marks:50, Time: 180 Min

Q1A. Design the amplifier shown in Fig. Q1A, for a drain current of 1mA and a voltage gain of 0.8. Assume $\mu_n C_{ox} = 100\ \mu A/V^2$, $V_{TH} = 0.5$ V, $\lambda = 0$, $V_{DD} = 1.8$ V, and $R_G = 50K\Omega$. (4)

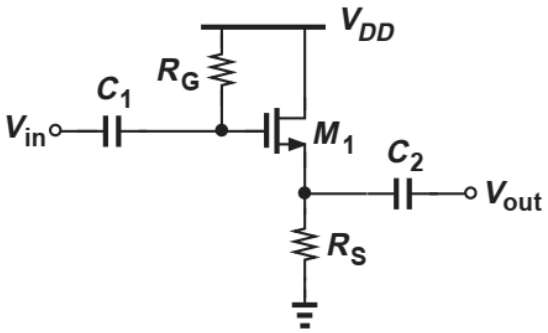


Fig. Q1A

Q1B. Calculate the bias current for the circuit shown in Fig. Q1B. If the gate voltage increases by 5mV, calculate the change in the drain voltage. Assume $\lambda = 0$, V_{TH}=0.5V, $\mu_n C_{ox}$= 100μA/V². (3)

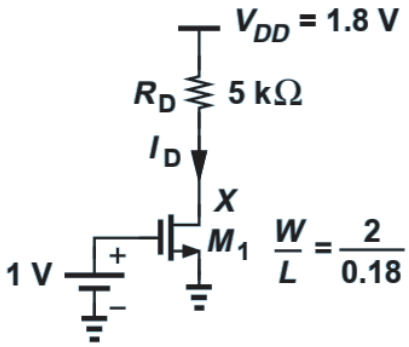


Fig. Q1B

Q1C. The circuit shown in Fig. Q1C incorporates the following device parameters: $(W/L)_{1,2} = 30$, $(W/L)_{3,4} = 40$, $I_{DS1} = I_{DS2} = I_{DS3} = I_{DS4} = 0.5$ mA. If $\mu_n C_{ox} = 100\ \mu A/V^2$, $\mu_p C_{ox} = 50\ \mu A/V^2$, $\lambda_n = 0.1\ V^{-1}$ and $\lambda_p = 0.15\ V^{-1}$, determine the voltage gain. (3)

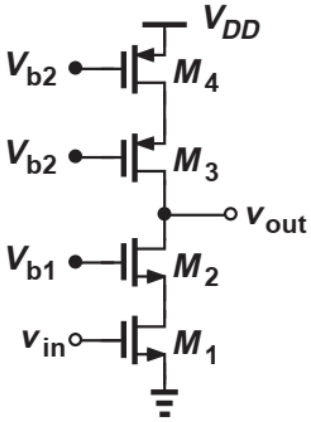


Fig. Q1C

Q2A. Design the circuit shown in Fig. Q2A, with a voltage gain of 20, power budget of 1 mW and $V_{DD} = 1.8$ V. Assume M_1 operates at the edge of saturation if the input common-mode level is 1 V. Also, $\mu_n C_{ox} = 2\mu_p C_{ox} = 100\ \mu A/V^2$, $V_{TH,n} = 0.5$ V, $V_{TH,p} = -0.4$ V, $\lambda_n = 0.5\lambda_p = 0.1\ V^{-1}$. (4)

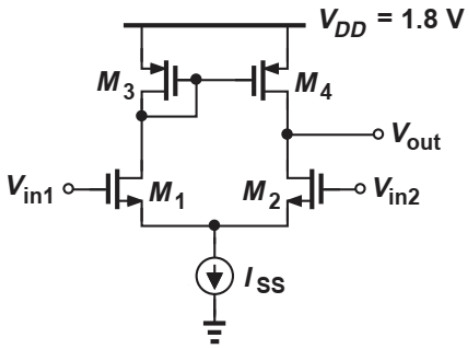


Fig. Q2A

Q2B. An integrated circuit shown in Fig. Q2B, employs the source follower and the common-source stage, as shown below. Design a current mirror that produces I_1 and I_2 from a 0.3mA reference. (3)

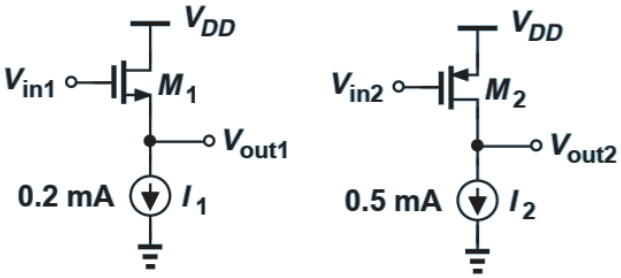


Fig. Q2B

Q2C. For the circuit shown in Fig. Q2C, the voltage drop across R_S is 200 mV. Calculate the minimum allowable value of W/L if M_1 remains in saturation. Also, determine the values of R_1 and R_2 to obtain an input impedance of at least 30KΩ?. Assume $\mu_n C_{ox} = 100\ \mu A/V^2$, V_{TH}=0.5V. (3)

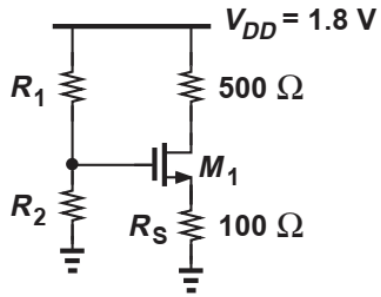


Fig. Q2C

Q3A. Calculate the single-ended and double-ended differential voltage gain for the circuits shown in Fig. Q3A. Assume perfect symmetry and $\lambda \neq 0$. (4)

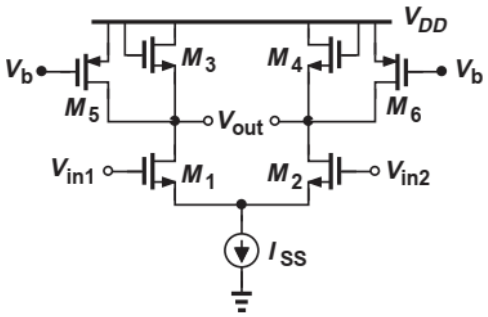


Fig. Q3A

Q3B. Determine the polarity of feedback for the circuits shown in Fig. Q3B. Explain the steps involved in the analysis. (3)

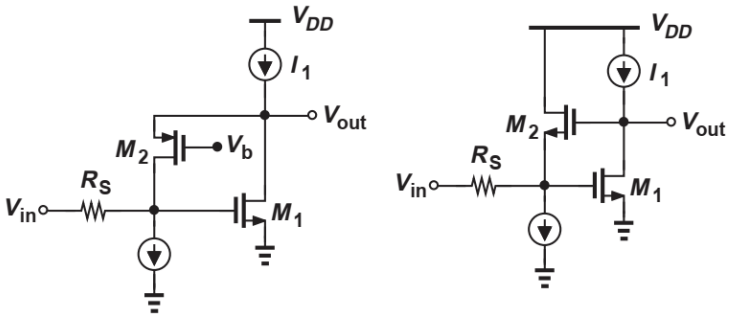


Fig. Q3B

Q3C. For the circuit shown in the Fig. Q3C, obtain the expressions for gain and frequency of oscillation. Also, extend the result for Hartley and Colpitt's oscillator. (3)

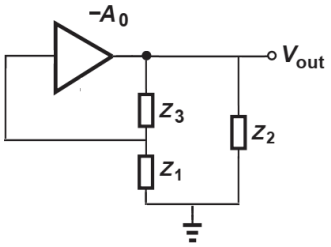


Fig. Q3C

Q4A. Determine the closed-loop gain, closed-loop input impedance, and closed-loop output impedances for the circuit shown in Fig. Q4A. Consider I/O effects (Loading Effect). Assume $\lambda=0$ for both M_1 and M_2. (4)

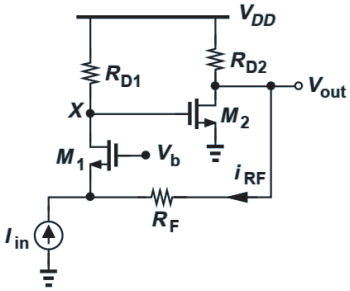


Fig. Q4A

Q4B. Design a circuit using a single OPAMP to obtain the following output.

$V_O = 2V_1 + 4V_2 - 3V_3 - 6V_4$ where V_1, V_2, V_3 and V_4 are the input, (Take R_F=12KΩ) (3)

Q4C. The circuit shown in Fig. Q4C has $V_{IN} = 5V$, R = 10KΩ, $\pm V_{sat} = \pm 12$ V, and a resistive load R_L. Find (i) i_O (ii) voltage compliance; (iii) maximum permissible value of R_L. (3)

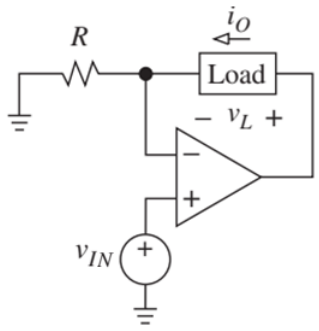


Fig. Q4C

Q5A. Design a 4th order Butterworth LPF with f_c =1KHz and passband gain of 2.575. (Take C=0.1μF) (4)

Q5B. Analyze the circuit shown in Fig. Q5B, determine the upper threshold and lower threshold point and plot the voltage transfer characteristics. (3)

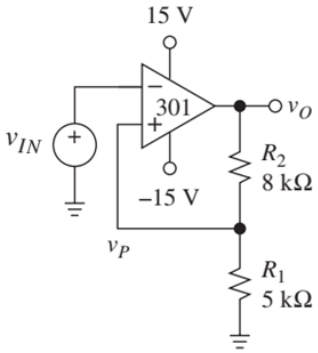


Fig. Q5B

Q5C. If V_{cc} = 6V and V_I=3+3Cosπt. Determine & sketch the output of the circuit shown in Fig. Q5C. (3)

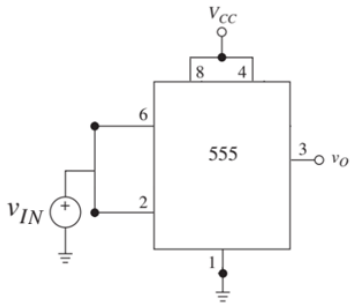


Fig. Q5C